

AI Search Framework: Executive Summary & Scientific Findings

Authoritative Briefing on the All-SLM Architecture vs. Multi-Scale Baselines | September 21, 2026

Definitive Research Finding: With all architectural bugs, single-node decomposer collapse, generative aggregator drift, and output truncation artifacts systematically diagnosed and resolved, **small language models ($\leq 5B$) hit an insurmountable parametric capacity ceiling on compound multi-domain engineering tasks against $\geq 32B$ frontier baselines.** Across 24 double-blind trials in Step 6 and 12 trials in Step B: **0.0% Win Rate vs $\geq 32B$ baselines** on un-truncated trials.

Essential Context on the $\sim 20B$ Tier: The SLM pipeline's wins vs $\sim 20B$ occurred solely when the 20B model entered unprompted infinite token repetition loops on RFC citations. When the 20B baseline generated coherent text without looping, the 20B model achieved a **100% win rate (6/6 trials)** and the SLM won **0.0%**.

Step B Audited Findings & General Rule Truncation Diagnostics:

- Authentic Audited Gain on $V3_CD_01$ (+0.166 pts / +3.3%):** Multi-turn execution feedback with sandboxed traceback injection raised composite quality from 1.667 (33.3%) to **1.833 (36.7%)** on clean, un-truncated trials vs 32B and 120B.
- Voiding of $V3_CD_41$ Truncation Delta:** All 5 SLM wins on $V3_CD_41$ coincided with comparator baselines cutting off mid-code. Under the General Rule, these trials are completely voided.
- No-Blending Rule:** Blending $V3_CD_01$ with the voided $V3_CD_41$ score (which would yield a spurious 62.2% pair score) is strictly prohibited. Across Step A and Step B, the only reproducible gain is the +0.166 pt coding improvement.

Multi-Scale Baseline Ladder (Step 6 Benchmark)

Baseline Tier	Model	Host Platform	Win Rate	Swap Agreement	Core Finding / Differentiator
Tier 1 ($\sim 20B$)	openai/gpt-oss-20b	Groq LPU	25.0% (2/8)	100.0% (audited)	Wins came solely from 20B infinite token loop on 1 query; 0% (0/6) when 20B was functional
Tier 2 ($\sim 32B$)	gemini-2.5-flash	Google AI Studio	0.0% (0/8)	100.0%	Frontier mathematical rigor and continuous stencil superiority
Tier 3 (120B)	openai/gpt-oss-120b	Groq LPU	0.0% (0/8)	100.0%	Monolithic multi-variable synthesis superiority

Chronological Progression Across Research Phases

- v1-v2:** Established 180-query stratified benchmark; audited and eradicated synthetic score imputation and proxy shortcuts; enacted Hard Rule 13 (distinct-roster preflight).
- v3:** Deployed Vulkan SLMs on RTX 3050; 9.4% win rate; isolated verbatim specialist confabulations (RFC mislabeling, phantom data.csv).
- v4 (Steps 1-5):** Proved isolated tools succeed (Retrieval Grounding: 100% win rate; Mechanical Code Verification: 66.7% win rate), but composite pipeline collapsed (0%) due to 3B aggregator domain drift and single-node decomposer collapse. Enacted Hard Rule 15.
- Step 6:** Deployed Deterministic Template Aggregation, eliminating generative drift. Audited against 3-tier scaling ladder.
- Step A & Step B:** Expanded grounding (+0.333 pts on Step A pair); introduced multi-turn execution retry feedback (+0.166 pts on clean $V3_CD_01$). Voided truncation artifacts on $V3_CD_41$.

Actionable Decision Framework for Mentor Review

- Pathway A (Accept Parametric Ceiling - Recommended):** Formally close the study as a rigorous empirical negative result demarcating the limits of $\leq 5B$ models on compound engineering tasks against $\geq 32B$ baselines and functioning 20B models.
- Pathway B (Request Authorization to Relax $\leq 5B$ Constraint to Test 7B-8B):** Testing genuine 7B-8B models (Llama-3.1-8B, Qwen-2.5-7B) is an explicit request to relax the mentor's operative $\leq 5B$ constraint, requiring formal new mentor authorization and cloud compute credits to test if 8B capacity bridges the gap to 32B.
- Pathway C (Isolated $\leq 5B$ Aggregator Test):** Test a stronger $\leq 5B$ model (e.g. Qwen-2.5-3B) as aggregator alone, isolating aggregation quality from specialist generation.